Predictive Analytics: Problem Set 7

Diyasha Basu, 707

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Shrinkage Methods : Comparison of least squares, ridge regression, LASSO and Elastic Net

```
data(Hitters)
Hitters <- na.omit(Hitters)
```

1. Split the data set into a training set and a test set randomly keeping 100 observations in test set and remaining in the training set.

```
set.seed(123)
test_idx  = sample(1:nrow(Hitters), 100)
train7    = Hitters[-test_idx, ]
test7     = Hitters[test_idx, ]

X_train = model.matrix(Salary ~ ., train7)[, -1]
y_train = train7$Salary
X_test  = model.matrix(Salary ~ ., test7)[, -1]
y_test  = test7$Salary

cat("Train size:", nrow(train7), "| Test size:", nrow(test7), "\n")

## Train size: 163 | Test size: 100
```

2. Train the model using least squares and report the train MSE and test MSE.

```
ols_fit   = lm(Salary ~ ., data = train7)
ols_train = predict(ols_fit, train7)
ols_test = predict(ols_fit, test7)

ols_train_mse = mean((y_train - ols_train)^2)
ols_test_mse = mean((y_test  - ols_test)^2)

cat("OLS Train MSE:", round(ols_train_mse, 2), "\n")

## OLS Train MSE: 105334.2

cat("OLS Test MSE: ", round(ols_test_mse,  2), "\n")

## OLS Test MSE:  87803.91
```

3. Fit ridge regression models on the training set, with grid values of λ. Report the following:

(i) the dimension of the coefficient matrix so obtained.

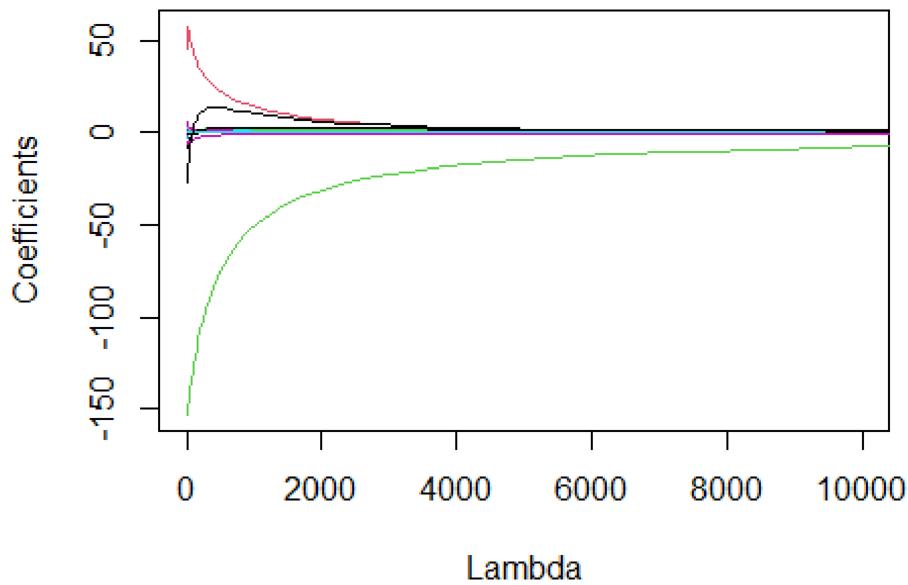
(ii) The l2 norm of the regression coefficients corresponding to the 50th and 60th choice of λ. Comment on the output.

(iii) Test MSE corresponding to these two choices of λ.

```
# Grid of lambda values
lambda_grid = 10^seq(5, -2, length = 100)

ridge_fit = glmnet(X_train, y_train, alpha = 0, lambda = lambda_grid)

# (i) Dimension of coefficient matrix
cat("Coefficient matrix dimensions:", dim(coef(ridge_fit)), "\n")

## Coefficient matrix dimensions: 20 100

# (ii) L2 norm at 50th and 60th lambda
l2_50 = sqrt(sum(coef(ridge_fit)[-1, 50]^2))
l2_60 = sqrt(sum(coef(ridge_fit)[-1, 60]^2))
cat("L2 norm at lambda[50] =", round(lambda_grid[50], 4), ":", round(l2_50,
4), "\n")

## L2 norm at lambda[50] = 34.3047 : 155.9135

cat("L2 norm at lambda[60] =", round(lambda_grid[60], 4), ":", round(l2_60,
4), "\n")

## L2 norm at lambda[60] = 6.7342 : 165.5575

cat("Larger lambda => smaller L2 norm (more shrinkage)\n")

## Larger lambda => smaller L2 norm (more shrinkage)

# (iv) Test MSE for 50th and 60th lambda
pred_50 = predict(ridge_fit, s = lambda_grid[50], newx = X_test)
pred_60 = predict(ridge_fit, s = lambda_grid[60], newx = X_test)
cat("Test MSE at lambda[50]:", round(mean((y_test - pred_50)^2), 2), "\n")

## Test MSE at lambda[50]: 89317.28

cat("Test MSE at lambda[60]:", round(mean((y_test - pred_60)^2), 2), "\n")

## Test MSE at lambda[60]: 87580.18
```

4. Make a line plot (for all the predictors) of the standardized coefficients against the grid value of λ chosen by you. Interpret the graph.

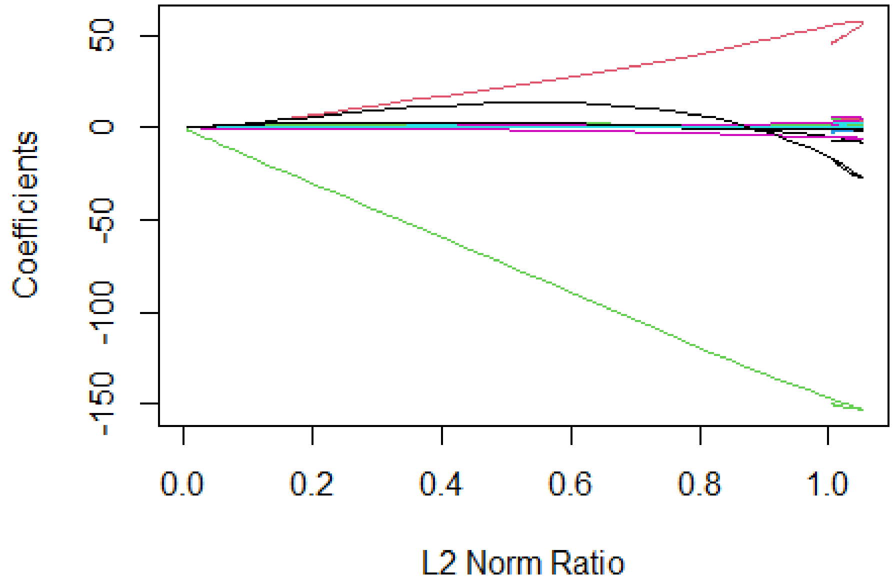
```
beta = as.matrix(ridge_fit$beta)
lambda = ridge_fit$lambda

matplot(lambda, t(beta), type = "l", lty = 1,
        xlab = "Lambda", ylab = "Coefficients",xlim = c(1e-2, 1e4))
```



Interpretation: As λ increases, all coefficients shrink toward zero. No coefficient reaches exactly zero — Ridge retains all predictors but with reduced magnitudes.

5. How would the graph in (4) change if instead of λ you plot on the x axis the ratio of l2 norm of the ridge regression coefficient to the least square regression coefficient.

```
ridge_coefs = predict(ridge_fit, type = "coefficients")

# Compute L2 norm for each Lambda
ridge_l2_norm = apply(ridge_coefs[-1, ], 2, function(b) sqrt(sum(b^2)))

# Least squares coefficients
lm_model = lm(y_train ~ X_train)
lm_coefs = coef(lm_model)[-1]
lm_l2_norm = sqrt(sum(lm_coefs^2))

# Ratio
ratio = ridge_l2_norm / lm_l2_norm

# Plot coefficients vs ratio
matplot(ratio, t(ridge_coefs[-1, ]), type = "l", lty = 1,
        xlab = "L2 Norm Ratio", ylab = "Coefficients")
```



6. Run the Ridge regression on training data by choosing λ by cross-validation. Find the test MSE for the optimal choice of λ obtained using cross validation.

```
set.seed(42)
cv_ridge = cv.glmnet(X_train, y_train, alpha = 0)
best_lambda_ridge = cv_ridge$lambda.min
cat("Best lambda (CV):", round(best_lambda_ridge, 4), "\n")

## Best lambda (CV): 368.4021

ridge_pred_cv = predict(cv_ridge, s = best_lambda_ridge, newx = X_test)
ridge_test_mse = mean((y_test - ridge_pred_cv)^2)
cat("Ridge Test MSE (CV lambda):", round(ridge_test_mse, 2), "\n")

## Ridge Test MSE (CV lambda): 93231.02
```

7. Fit a LASSO model on the training set, with λ chosen by cross-validation. Report the test MSE, along with the number of non-zero coefficient estimates.

```
set.seed(42)
cv_lasso = cv.glmnet(X_train, y_train, alpha = 1)
best_lambda_lasso = cv_lasso$lambda.min
cat("Best lambda (CV):", round(best_lambda_lasso, 4), "\n")

## Best lambda (CV): 24.2384

lasso_pred = predict(cv_lasso, s = best_lambda_lasso, newx = X_test)
lasso_test_mse = mean((y_test - lasso_pred)^2)
```

```r
lasso_coefs = coef(cv_lasso, s = best_lambda_lasso)
nonzero = sum(lasso_coefs != 0) - 1  # exclude intercept
cat("LASSO Test MSE:", round(lasso_test_mse, 2), "\n")

## LASSO Test MSE: 95172.6

cat("Number of non-zero coefficients:", nonzero, "\n")

## Number of non-zero coefficients: 8
```

8. Fit an Elastic Net model on the training set, with λ1 and λ2 chosen by cross-validation. Report the test MSE. Also report the predictors with non-zero coefficients.

```r
alphas = seq(0.1, 0.9, by = 0.1)
cv_errors = sapply(alphas, function(a) {
  cv_fit = cv.glmnet(X_train, y_train, alpha = a)
  min(cv_fit$cvm)
})

best_alpha = alphas[which.min(cv_errors)]
cat("Best alpha:", best_alpha, "\n")

## Best alpha: 0.2

set.seed(42)
cv_enet     = cv.glmnet(X_train, y_train, alpha = best_alpha)
best_lam_en = cv_enet$lambda.min
cat("Best lambda (CV):", round(best_lam_en, 4), "\n")

## Best lambda (CV): 100.6157

enet_pred = predict(cv_enet, s = best_lam_en, newx = X_test)
enet_test_mse = mean((y_test - enet_pred)^2)

enet_coefs = coef(cv_enet, s = best_lam_en)
nonzero_en = sum(enet_coefs != 0) - 1
cat("Elastic Net Test MSE:", round(enet_test_mse, 2), "\n")

## Elastic Net Test MSE: 94026.38

cat("Non-zero predictors:", nonzero_en, "\n")

## Non-zero predictors: 12

cat("Non-zero predictor names:\n")

## Non-zero predictor names:

print(rownames(enet_coefs)[enet_coefs[,1] != 0 & rownames(enet_coefs) != "(In
tercept)"])
```

```
##  [1] "Hits"      "HmRun"     "RBI"       "Walks"     "CAtBat"    "CHits"    

##  [7] "CHmRun"    "CRuns"     "CRBI"      "LeagueN"   "DivisionW" "PutOuts"
```

9. Compare the test MSE 's of least squares regression, Ridge regression, LASSO and ELASTIC NET and comment on how accurately we can predict the Salary of a baseball player.

```
comparison = data.frame(
  Method     = c("OLS", "Ridge (CV)", "LASSO (CV)", "Elastic Net (CV)"),
  Test_MSE   = round(c(ols_test_mse, ridge_test_mse, lasso_test_mse, enet_tes
t_mse), 2)
)
print(comparison)
```

```
##             Method Test_MSE
## 1              OLS 87803.91
## 2       Ridge (CV) 93231.02
## 3       LASSO (CV) 95172.60
## 4 Elastic Net (CV) 94026.38
```

Interpretation: Shrinkage methods (Ridge, LASSO, Elastic Net) typically outperform OLS on test data because they reduce overfitting. LASSO also performs variable selection (many zero coefficients), making the model more interpretable. Elastic Net combines Ridge and LASSO penalties and can outperform either alone. The method with the lowest test MSE gives the most accurate salary predictions.